

Exhibit 4-6.*Level 2 analysis of assessment data.***Attainment of Student Outcome #1****An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.**

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	67.2	67.8	66.9	70.9	69.6	73.2
% (2 = Developing)	19.3	17.8	23.2	18.8	20.9	18.6
% (1 = Unsatisfactory)	13.4	14.4	9.9	10.3	9.5	8.2
% > Developing	86.6	85.6	90.1	89.7	90.5	91.8
Average Rating	2.51	2.48	2.52	2.57	2.37	2.59

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		61.7	71.0	62.7	49.5	56.5
% (2 = Developing)		27.1	22.6	30.0	41.4	35.3
% (1 = Unsatisfactory)		11.3	6.3	7.3	9.1	8.2
% > Developing		88.8	93.7	92.7	90.9	91.8
Average Rating		2.40	2.59	2.46	2.29	2.48

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

*Note different rating system than above Methods #1- #3:**(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)*

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.47				1.50	1.44

Attainment of Student Outcome #2

An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		82.5	80.1	84.0	55.6	64.4
% (2 = Developing)		14.5	12.9	13.1	19.4	24.0
% (1 = Unsatisfactory)		2.9	7.0	2.8	25.0	11.5
% > Developing		97.1	93.0	97.2	75.0	88.5
Average Rating		2.75	2.73	2.81	1.75	2.25

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		78.3	71.4	64.6	56.4	63.0
% (2 = Developing)		18.3	27.0	32.3	38.2	30.4
% (1 = Unsatisfactory)		3.3	1.6	3.1	5.5	6.5
% > Developing		96.7	98.4	96.9	94.5	93.5
Average Rating		2.70	2.70	2.62	2.51	2.57

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	64.9	82.7	72.8			76.2
% (2 = Developing)	34.4	17.3	27.2			23.8
% (1 = Unsatisfactory)	0.7	0.0	0.0			0.0
% > Developing	99.3	100.0	100.0			100.0
Average Rating	2.64	2.83	2.73			2.74

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.61				1.62	1.60

Attainment of Student Outcome #3**An ability to communicate effectively with a range of audiences.**

Assessment Method #1 (Direct / Formative): Course Rubrics

Oral

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		75.0	82.6			77.4
% (2 = Developing)		23.4	15.7			22.6
% (1 = Unsatisfactory)		1.6	1.7			0.0
% > Developing		98.4	98.5			100.0
Average Rating		2.72	2.75			2.63

Written

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		88.6	76.9	76.6	70.0	73.5
% (2 = Developing)		8.4	18.8	22.2	29.3	25.8
% (1 = Unsatisfactory)		3.1	4.3	1.2	0.8	0.7
% > Developing		96.9	95.7	98.8	99.3	99.3
Average Rating		2.86	2.73	2.75	2.69	2.72

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	76.2	84.0	82.8			92.9
% (2 = Developing)	22.7	16.0	16.4			7.1
% (1 = Unsatisfactory)	1.1	0.0	0.8			0.0
% > Developing	98.9	100.0	99.2			100.0
Average Rating	2.75	2.84	2.82			2.90

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

*Note different rating system than above Methods #1- #3:**(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)*

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.40				1.45	1.46

Attainment of Student Outcome #4

An ability to recognize ethical, safety, and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

Assessment Method #1 (Direct / Formative): Course Rubrics

Safety and Environment (S&E)

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	100.0	90.1	99.2	89.7	100.0	100.0
% (2 = Developing)	0.0	8.89	0.8	9.5	0.0	0.0
% (1 = Unsatisfactory)	0.0	1.1	0.0	0.8	0.0	0.0
% > Developing	100.0	98.9	100.0	99.2	100.0	100.0
Average Rating	3.00	2.90	2.99	2.89	3.00	3.00

Global, economic, societal, ethics (GESE)

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	90.3	63.1	88.5	65.6		
% (2 = Developing)	6.5	36.9	1.8	29.5		
% (1 = Unsatisfactory)	3.2	0.0	9.8	4.9		
% > Developing	96.8	100.0	90.2	95.1		
Average Rating	2.87	2.62	2.79	2.61		

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		82.8	83.1	76.9	69.1	57.2
% (2 = Developing)		13.9	16.9	22.1	23.6	26.8
% (1 = Unsatisfactory)		3.3	0.0	1.0	7.3	15.9
% > Developing		96.7	100.0	98.5	92.7	84.1
Average Rating		2.71	2.74	2.65	2.52	2.41

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.30				1.26	1.50

Attainment of Student Outcome #5

An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		90.5	66.6	83.2	88.7	81.5
% (2 = Developing)		8.5	31.8	16.3	7.9	16.1
% (1 = Unsatisfactory)		1.0	1.6	0.4	3.4	2.4
97.6% > Developing		99.0	98.4	99.6	96.6	97.6
Average Rating		2.94	2.87	2.90	2.85	2.75

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		93.3	81.0	89.2	89.1	65.2
% (2 = Developing)		6.7	14.3	9.2	10.9	32.6
% (1 = Unsatisfactory)		0.0	4.8	1.5	0.0	2.2
% > Developing		100.0	95.2	98.5	100.0	97.82
Average Rating		2.70	2.56	2.67	2.71	2.63

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.45				1.39	1.60

Attainment of Student Outcome #6

An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		73.9	79.3	66.9	66.7	69.6
% (2 = Developing)		25.2	19.0	33.1	31.9	29.0
% (1 = Unsatisfactory)		0.9	1.7	0.0	1.4	1.3
% > Developing		99.1	98.3	100.0	98.6	98.7
Average Rating		2.72	2.76	2.72	2.69	2.68

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.54				1.63	1.54

Attainment of Student Outcome #7

An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		84.1	70.6	53.0	89.3	96.0
% (2 = Developing)		14.3	21.1	47.0	11.0	4.0
% (1 = Unsatisfactory)		1.6	8.3	0.0	0.0	0.0
% > Developing		98.4	91.7	100.0	100.0	100.0
Average Rating		2.83	2.62	2.53	2.89	2.96

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		70.0	74.6	86.2	67.3	78.3
% (2 = Developing)		30.0	25.4	12.3	23.6	21.7
% (1 = Unsatisfactory)		0.0	0.0	1.5	9.1	0.0
% > Developing		100.0	100.0	98.5	90.9	100.0
Average Rating		2.50	2.56	2.63	2.44	2.76

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.51				1.57	1.75